

Project P10 Research Notebook

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize
Theta-Gamma Phase-Amplitude Coupling in Alzheimer's Disease

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Project P10 Daily Research Notebook

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize Theta-Gamma Coupling in Alzheimer's Disease

Researcher: Amaar Chughtai **School:** Valley Christian High School **Fair:** Synopsys Science and Engineering Fair 2026 **Research Period:** December 10, 2025 – March 3, 2026

December 10, 2025: Project Kickoff and Initial Setup

Plan: Begin project setup, download EEG dataset, establish development environment.

Goal: Get basic data loading working and understand the dataset structure.

What I Did:

- Downloaded OpenNeuro ds005048 dataset (847 MB) - "40 Hz Auditory Entrainment in Dementia"
- Set up Python environment with PyTorch, MNE-Python, h5py
- Initial attempt at data loading failed - MATLAB v7.3 files not readable with standard MNE tools
- Discovered dataset uses HDF5 format (.set files) with companion binary data (.fdt files)

Results:

- Dataset confirmed: 35 elderly dementia patients, 19 EEG channels @ 250 Hz
- Stimulation protocol: 40s ON / 20s OFF auditory pulses (40 Hz, 5 kHz carrier)
- Data loading blocked - need custom loader

Problems: Standard Python EEG libraries can't read MATLAB v7.3 HDF5 format.

December 12, 2025: Custom Data Loader Development

Plan: Build custom loader for MATLAB HDF5 files.

Goal: Successfully load EEG data and extract epoch boundaries.

What I Did:

- Developed custom loader using h5py for .set metadata + numpy for .fdt binary data
- **Critical discovery:** MATLAB stores matrices in Fortran order (column-major)
- Used order='F' in reshape - without this, data was silently transposed
- Extracted stimulus/rest boundaries from BIDS events.tsv files

Code snippet:

```
# Read .set HDF5 metadata
with h5py.File(set_path, 'r') as f:
    n_channels = int(f['nbchan'][0, 0])
```

```
n_samples = int(f['pnts'][0, 0])

# Read binary EEG data (CRITICAL: order='F')
data = np.fromfile(fdt_path, dtype=np.float32)
data = data.reshape((n_channels, n_samples), order='F')
```

Results:

- Successfully loaded all 35 subjects
- Total windows: 17,283 (2-second windows, 50% overlap)
- Channel selection: 7 frontal channels (Fp1, Fp2, F7, F3, Fz, F4, F8)

December 15, 2025: PAC Computation and Preprocessing

Plan: Implement Phase-Amplitude Coupling (PAC) calculation using Modulation Index.

Goal: Extract PAC labels from EEG data for each 2-second window.

What I Did:

- Implemented Modulation Index (Tort et al. 2010): θ phase (4-8 Hz) $\times$ γ amplitude (38-42 Hz)
- Applied light preprocessing: 0.5-80 Hz bandpass, 50 Hz notch, ± 100 μ V artifact rejection
- Computed PAC at epoch level (20-40s blocks), then assigned to constituent windows

PAC Algorithm:

1. Bandpass filter: θ (4-8 Hz), γ (38-42 Hz)
2. Hilbert transform phase/amplitude
3. Bin phase space into 18 bins (20° each)
4. MI = KL divergence from uniform distribution

Results:

Metric	Value
PAC range	[0.000006, 0.000701]
PAC mean	0.000044
PAC std	2.2×10
Train/Val/Test	11,736 / 2,725 / 2,822 windows
Subject split	24 / 5 / 6 subjects

December 18, 2025: EEGNet Baseline Model (V1)

Plan: Implement EEGNet architecture adapted for PAC regression.

Goal: Establish baseline performance for static PAC prediction.

What I Did:

- Adapted Lawhern et al. (2018) EEGNet from classification to regression
- Architecture: temporal conv depthwise spatial conv separable conv FC head
- Training: MSE loss, Adam optimizer, z-score normalized PAC targets

Architecture Details:

Block	Layer	Output Shape
Block 1	Conv2d temporal (8 filters, k=64)	(B, 8, 7, 500)
	Depthwise spatial (16 filters, k=7×1)	(B, 16, 1, 500)
	AvgPool2d, Dropout(0.5)	(B, 16, 1, 125)
Block 2	Separable conv	(B, 16, 1, 15)
Head	Flatten + Linear	(B, 1)

Results:

- **Test $R^2 = 0.287$**
- Parameters: 1,457
- Inference time: <1 ms on Apple Silicon

This became our static PAC predictor baseline.

December 22, 2025: Architecture Exploration (V2-V8)

Plan: Test multiple architectures to improve beyond $R^2 = 0.287$.

Goal: Find if model capacity is the bottleneck.

What I Did:

- V2: EEGNetV2 for delta-PAC prediction $R^2 = 0.06$ (failed)
- V3: SpecTempNet (180k params) initially $R^2 = 0.69$ (suspicious!)
- Discovered feature leakage: PAC-derived features in input were circular
- V4: ViT-TCNet (1.1M params) $R^2 = 0.252$ (overfitting)
- V5: Ridge Regression (200 params) $R^2 = 0.287$ (matches EEGNet!)
- V6-V8: Various CNNs, attention models all converged near 0.287

Key Discovery: PAC features from `pac_features.py` directly encode the target (correlation > 0.7).
 Ridge analysis: PAC features carried 96.6% of model weight = data leakage!

Results Table:

Version	Architecture	Parameters	Test R^2	Notes
V1	EEGNet	1,457	0.287	Baseline
V3	SpecTempNet	180,000	0.236	After leak fix

Version	Architecture	Parameters	Test R^2	Notes
V4	ViT-TCNet	1,100,000	0.252	Overfit
V5	Ridge	~200	0.287	Matches best

Conclusion: $R^2 = 0.287$ is the data ceiling for static prediction, not a model limitation.

January 8, 2026: Temporal Prediction Pivot

Plan: Since static prediction hit a ceiling, pivot to temporal forecasting.

Goal: Predict future PAC from sequences instead of improving instantaneous prediction.

What I Did:

- Reframed problem: predict PAC 5-10 seconds ahead from 20-second history
- Built 73-dimensional feature vectors per timestep:
 - 61 spectral features (5 bands $\times$ 7 channels + coherence)
 - 7 PAC-derived features (current, moving averages, differences)
 - 5 stimulation context features (state, timing, cycle phase)

Feature Engineering:

```
# Strictly causal PAC features - safe for temporal prediction
pac_ma2 = causal_moving_average(pac_current, window=2)
pac_diff1 = pac_current[t] - pac_current[t-1]
time_since_switch = seconds_since_stim_change
```

Results:

- Sequences: (batch, 20_timesteps, 73_features)
- Target: PAC at t+5 seconds
- Safe from leakage: predicting future PAC, not current

January 15, 2026: MultiscaleCausalTCN Development

Plan: Design temporal convolutional network for PAC forecasting.

Goal: Outperform persistence and linear baselines at 5+ second horizons.

What I Did:

- Built MultiscaleCausalTCN with causal depthwise-separable convolutions
- Dilation pattern [1, 2, 4, 8] receptive field = 31 timesteps
- GroupNorm for cross-subject stability, attention pooling, dual-head output

Architecture:

Component	Details	Parameters
Input projection	Linear(7364) + LayerNorm + SiLU	4,864
TCN blocks × 4	Causal DepthwiseConv1d + PointwiseConv1d	18,432
Attention pool	Conv1d(641) + softmax weighting	65
Dual heads	Future PAC + delta-PAC prediction	8,450
Total		31,043

Critical Design:

- Causal padding: `F.pad(x, ((kernel_size-1)*dilation, 0))`
- No future leakage: model cannot access $t+1$, $t+2$, etc.
- GroupNorm vs BatchNorm: stable across different patient baselines

Training:

- Loss: Huber (delta=1.0) for outlier robustness
- Optimizer: AdamW (lr=0.001, weight_decay=0.001)
- Early stopping: patience=20 epochs on validation R^2

January 22, 2026: Horizon Sweep Analysis

Plan: Test TCN performance across prediction horizons 1-10 seconds.

Goal: Find where TCN provides value over simpler baselines.

What I Did:

- Trained separate models for each horizon (1s, 2s, 3s, 5s, 8s, 10s)
- Compared against persistence ($y_{\text{future}} = y_{\text{current}}$) and Ridge regression
- Used target_smooth_window=5 for fair comparison across all models

Results:

Horizon	TCN R^2	Persistence R^2	Ridge R^2	TCN Advantage
1s	0.735	0.760	0.812	-0.025 (loses)
2s	0.470	0.488	0.542	-0.018 (loses)
3s	0.277	0.234	0.253	+0.043 (wins)
5s	0.254	-0.267	-0.393	+0.521
8s	0.240	-0.276	-0.211	+0.516
10s	0.278	-0.256	-0.212	+0.534

Key Finding:

- At 1-2s: PAC changes slowly, persistence works well

- At 3s: Crossover point where TCN starts winning
- At 5-10s: Baselines collapse (negative R^2), TCN maintains $R^2 \approx 0.25$
- Margin = +0.5 R^2 at clinically relevant horizons

This is the TCN's value proposition!

January 28, 2026: Closed-Loop Controller Integration

Plan: Integrate TCN into real-time controller framework.

Goal: Build complete closed-loop system with personalization.

What I Did:

- Built two-stage pipeline: EEGNet (current PAC) TCN (future PAC)
- Added PersonalizationModule: 30-second rolling baseline for z-score normalization
- Implemented decision logic with hysteresis (5-second minimum state duration)

System Architecture:

EEG window → EEGNet → current PAC → feature extraction (73D) →
 TCN → predicted future PAC → PersonalizationModule → z-score →
 decision logic → STIMULATE/REST/MAINTAIN

Decision Rules:

Condition	Action	Logic
$z < -0.5$	STIMULATE	PAC below baseline
$z > +0.5$	REST	PAC above baseline
$-0.5 \leq z \leq +0.5$	MAINTAIN	Stay in current state

TCN Predictive Extension:

- If predicted delta-PAC < -0.3 STIMULATE (proactive)
- If predicted delta-PAC $> +0.3$ REST (proactive)
- Otherwise fall back to reactive z-score logic

February 5, 2026: Real-Data Validation

Plan: Validate all controllers on the full 35-subject dataset.

Goal: Quantify TCN controller advantage on real EEG recordings.

What I Did:

- Offline replay analysis: each controller makes decisions on real EEG windows
- Compared 6 strategies: Fixed Schedule, Reactive, TCN Predictive, Hybrid, PI, Oracle
- Computed alignment metrics: sensitivity, specificity, PAC targeting gap

Validation Results (N=35 subjects):

Controller	Alignment	Low-PAC Stim	High-PAC Rest	PAC Gap (×10)
Fixed Schedule	45.0%	61.4%	28.6%	-6.6
Reactive	64.5%	51.7%	77.3%	+21.1
TCN Predictive	72.1%	82.6%	61.6%	+30.5
Oracle (theoretical)	100.0%	100.0%	100.0%	+33.3

Statistical Significance (TCN vs Reactive):

- Alignment: $g = 1.31$, $p < 0.001$ (large effect)
- Low-PAC targeting: $g = 4.47$, $p < 0.001$ (very large effect)
- PAC gap: $g = 1.57$, $p < 0.001$ (large effect)

Key Results:

1. **TCN achieved 72.1% vs 64.5% alignment** (8-point improvement)
2. **82.6% vs 51.7% therapeutic targeting** (60% improvement)
3. **92% of theoretical oracle performance** (30.5/33.3 PAC gap ratio)
4. **35/35 subjects benefited** (universal improvement)

February 12, 2026: Habituation Analysis

Plan: Analyze real-data evidence for neural habituation during 40 Hz stimulation.

Goal: Validate the core motivation for adaptive control.

What I Did:

- Analyzed PAC trajectories across stimulation blocks for all 35 subjects
- Compared first vs last stimulation block PAC levels
- Computed within-block habituation rates

Results:

Metric	Value
Population mean change	+4.5% (not significant, $p=0.542$)
Subjects declining (habituating)	17/35 (48.6%)
Subjects increasing (facilitating)	18/35 (51.4%)
Individual range	-66.8% to +149.1%

Key Insight: No population-level trend because cohort splits equally between habituators/facilitators. **This validates personalized adaptive control** - fixed schedules can't handle this heterogeneity.

February 18, 2026: Fatigue Model Sensitivity

Plan: Test adaptive vs fixed control across different fatigue assumptions.

Goal: Ensure results aren't artifacts of specific habituation models.

What I Did:

- Built EntrainmentSimulator with 4 fatigue models: exponential, step, heterogeneous, saturation
- Ran 360-second simulations with 10 trials per condition
- Swept fatigue rates from 0.000 (no fatigue) to 0.040 (severe)

Results:

Fatigue Rate	Fixed Efficiency	Adaptive Efficiency	Gain	p-value
0.000 (none)	5.381	5.401	+0.39%	0.492 (n.s.)
0.025 (high)	4.968	5.184	+4.35%	0.002
0.040 (severe)	4.819	5.092	+5.65%	0.010

Robust Across Fatigue Models:

Model	Improvement	Hedges' g	p-value
Exponential	+9.0%	2.31	<0.001
Saturation	+19.0%	3.66	<0.001
Heterogeneous	+8.9%	1.71	<0.001

Conclusion: Adaptive advantage grows with fatigue severity and holds across all models.

February 25, 2026: Final Validation and Robustness Testing

Plan: Comprehensive robustness analysis before submission.

Goal: Verify results aren't artifacts of parameter tuning or data splits.

What I Did:

- Threshold sweep: delta-z from 0.1 to 1.0
- Data integrity checks: leakage audits, causality verification
- Cross-validation stability analysis

Threshold Robustness:

Delta-z	Alignment	Low-PAC Stim	PAC Gap
0.2	68.5%	72.9%	26.3
0.3	73.7%	84.9%	32.4
0.5	73.7%	85.3%	33.8

Delta-z	Alignment	Low-PAC Stim	PAC Gap
1.0	73.8%	85.3%	34.0

Results plateau at delta-z ≥ 0.3 - performance stable across threshold choices.

Data Integrity Verification: No subject leakage between splits All features precede targets temporally Normalization fit on training data only Shuffle-label test confirms real signal ($R^2 = -0.332$)

March 1, 2026: Documentation and Submission Prep

Plan: Finalize documentation, poster, and submission materials.

Goal: Complete research notebook, poster board, and abstract.

What I Did:

- Compiled 22-page formal research notebook
- Created poster board for Synopsys fair
- Generated publication-quality figures
- Wrote 247-word abstract (within 250-word limit)

Final Project Statistics:

- **Codebase:** ~6,000 lines of Python
- **Models trained:** 20+ architectures tested
- **Data processed:** 35 subjects, 17,283 windows
- **Key result:** 72.1% vs 64.5% alignment, universal benefit across subjects

Submission Package:

1. Research notebook (22 pages)
2. Poster board (48x56)
3. Abstract (247 words)
4. Code repository with full reproducibility

March 3, 2026: Final Results Summary

Project Completed Successfully

Key Achievements:

1. **Temporal prediction breakthrough:** TCN maintains $R^2 = 0.24-0.28$ at 5-10s horizons where baselines fail (+0.5 R^2 margin)
2. **Real-data clinical validation:** 72.1% alignment vs 64.5% reactive control across all 35 subjects
3. **Therapeutic precision:** 82.6% vs 51.7% low-PAC targeting (60% improvement)

4. **Near-optimal performance:** 92% of theoretical oracle bound
5. **Universal benefit:** 35/35 subjects improved ($p < 0.001$)
6. **Robust across assumptions:** Holds for all fatigue models and threshold parameters

Technical Innovation:

- First predictive closed-loop controller for 40 Hz entrainment
- MultiscaleCausalTCN: 31k parameter efficient architecture
- Personalized z-score adaptation with rolling baselines
- Leakage-safe temporal feature engineering

Clinical Impact:

- Addresses 30% non-responder rate in current trials
- Reduces wasted stimulation while improving therapeutic targeting
- Validated on real EEG from dementia patients
- Ready for live closed-loop deployment

Personal Growth: This project taught me that breakthrough research often requires pivot points. When static PAC prediction hit a ceiling (8 architectures, all $R^2 \approx 0.287$), the solution wasn't bigger models—it was reframing the problem temporally. The real innovation came from asking "When should we stimulate?" instead of "How well can we predict PAC?"

Working with real EEG data from Alzheimer's patients made this personal. My grandmother's dementia motivated this work, and seeing individual patient heterogeneity in the data reinforced why personalized medicine matters.

The project is now ready for submission to Synopsys Science Fair 2026.

Equipment and Software Used

Hardware:

- MacBook Pro (Apple M1 Pro, 16GB RAM)
- Internet connection for dataset download

Software:

- Python 3.13.3
- PyTorch (MPS backend)
- NumPy, SciPy, scikit-learn
- h5py, MNE-Python
- Git version control

Dataset:

- OpenNeuro ds005048 v1.0.1
- 35 elderly dementia patients
- Tehran Memory Clinic, Iran
- 847 MB total download

Total Project Duration: 84 days (December 10, 2025 – March 3, 2026)

"The best way to predict the future is to create it." - This project created a future where 40 Hz therapy adapts to each patient's unique neural dynamics.